

GUAYAQUIL, Ecuador

WMO: 842030

Lat: 2.157S

Lon: 79.884W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	66.0	66.6	59.1	74.8	72.2	60.4	78.6	72.0	18.5	72.7	17.1	72.8	8.6	230	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.0	91.3	75.1	89.8	75.4	89.2	75.4	79.8	86.0	78.8	85.0	77.9	84.6	6.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.7	83.9	77.0	140.7	82.2	75.6	133.9	81.5	43.5	86.3	42.2	85.7	41.3	84.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.8	13.6	DB	65.6	94.1	3.4	1.5	63.1	95.2	61.2	96.1	59.3	96.9	56.8	98.0	(4)
(5)			WB	60.4	79.7	3.4	1.5	58.0	80.8	56.0	81.7	54.1	82.5	51.7	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	80.4	80.2	81.1	81.0	80.2	77.6	76.5	76.5	77.4	77.3	78.2	80.4	(6)	
	DBStd	2.99	2.55	2.39	2.22	2.11	2.38	2.70	2.78	2.54	2.57	2.19	2.26	2.47	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	10547	943	846	964	931	935	828	821	822	821	847	846	943	(10)	
	CDD65	5072	478	426	499	481	470	378	356	357	371	382	396	478	(11)	
	CDH74	40323	4286	3775	4786	4690	4228	2657	2191	2135	2400	2320	2694	4161	(12)	
	CDH80	11944	1331	1039	1491	1523	1317	665	489	515	684	610	792	1488	(13)	
(14) Wind	WSAvg	6.5	4.9	4.3	4.5	4.9	5.5	6.8	7.5	8.1	8.3	8.3	7.7	6.8	(14)	
(15) Precipitation	PrecAvg	34.1	7.9	9.8	10.9	6.3	2.0	0.1	0.0	0.0	0.1	0.1	0.0	1.3	(15)	
	PrecMax	81.1	27.6	29.6	35.9	23.8	8.2	1.1	0.3	0.8	0.8	0.7	0.2	11.6	(16)	
	PrecMin	13.0	0.6	1.7	0.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	17.3	6.2	7.1	8.0	5.5	2.7	0.2	0.0	0.1	0.2	0.2	0.0	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	91.1	91.3	91.3	90.9	89.5	89.3	88.8	89.9	89.8	90.0	93.0	(19)	
		MCWB	73.8	77.9	77.8	76.7	76.3	75.1	75.1	75.1	74.7	73.7	73.9	73.9	(20)	
	2%	DB	90.0	89.3	89.6	89.6	89.3	87.6	86.3	86.3	87.8	87.5	88.0	91.0	(21)	
		MCWB	74.7	76.9	77.1	76.4	76.4	74.5	74.3	73.8	73.9	73.2	73.5	73.5	(22)	
	5%	DB	88.1	87.6	88.0	88.1	87.8	85.8	84.5	84.5	86.0	85.7	86.3	89.3	(23)	
		MCWB	74.7	76.4	76.7	76.3	76.0	74.0	73.2	72.8	73.2	72.5	72.8	73.4	(24)	
	10%	DB	86.3	85.9	86.4	87.3	86.2	84.0	82.6	82.7	84.2	83.8	84.4	87.5	(25)	
		MCWB	74.4	76.1	76.3	76.1	75.5	73.5	72.4	72.0	72.3	71.8	72.0	73.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	82.0	82.1	79.6	79.2	77.1	77.4	77.1	77.4	75.3	79.1	76.9	(27)	
		MCDB	87.0	88.0	88.1	87.6	86.9	85.7	86.7	86.7	86.8	86.8	85.7	87.5	(28)	
	2%	WB	78.3	79.4	79.6	78.3	78.2	75.7	75.6	75.0	75.1	74.0	76.1	75.6	(29)	
		MCDB	83.0	84.6	85.6	86.2	85.8	84.3	84.8	84.8	85.5	84.9	83.0	86.2	(30)	
	5%	WB	76.9	78.2	78.4	77.4	77.2	74.7	74.0	73.6	73.8	72.9	74.2	74.6	(31)	
		MCDB	83.1	83.5	84.5	85.3	84.9	83.1	82.5	82.9	83.9	83.3	83.1	85.2	(32)	
	10%	WB	75.8	77.2	77.4	76.6	76.2	73.8	72.8	72.5	72.7	72.2	72.8	73.7	(33)	
		MCDB	82.7	82.6	83.8	84.4	83.9	82.1	80.9	81.5	82.5	82.1	82.8	84.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	10.8	11.4	12.0	12.0	12.3	12.4	13.5	14.5	13.7	13.9	14.0	(35)	
		MCDBR	14.4	12.8	12.9	13.5	13.7	14.3	14.4	15.5	16.3	16.3	16.2	16.6	(36)	
	5% WB	MCWBR	5.4	5.3	5.7	6.2	6.4	6.4	6.5	7.1	7.2	7.2	6.8	6.2	(37)	
		MCWBR	11.7	11.2	11.9	12.4	12.3	13.0	13.4	14.5	15.2	14.7	14.4	14.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.507	0.505	0.502	0.553	0.556	0.525	0.507	0.543	0.588	0.549	0.566	0.555	(40)	
		taud	2.157	2.179	2.192	2.027	2.007	2.100	2.162	2.049	1.921	2.033	1.965	2.002	(41)	
	Ebn at Noon		265	266	265	244	235	239	246	244	239	252	247	250	(42)	
		Edh at Noon	50	50	49	56	55	49	46	54	63	57	61	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1304	1321	1420	1446	1297	1159	1135	1210	1311	1279	1345	1349	(44)	
	RadStd		133	73	105	79	107	95	96	108	91	105	135	107	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page